

waves and particles

As the evidence mounted, light, which used to be thought of as a wave, had to be regarded as being both wave and particle. In the early 1920s, the French physicist Louis de Broglie came up with the inspired suggestion that electrons, which used to be thought of as particles, had to be regarded as being both particle and wave. In fact,



subatomic sculpture
By making a ring of atoms to act as a "corral," researchers in the 1990s were able to trap electrons as waves moving across the surface of a metal like ripples on the surface of a pond.

he said that everything had to be regarded as being both particle and wave, but that the amount of waviness associated with everyday objects, such as people or house plants, doesn't show up because they have so much mass.

Using Einstein's work on photons, an equation was discovered that linked the particle and wave properties of light.

The equation says that the wavelength associated with a photon multiplied by its momentum (the force that keeps an object moving) is equal to Planck's constant, h . De Broglie realized that the equation is universally true, so that a particle such as an electron (or anything else) that has a certain momentum must also have a certain wavelength, which can be calculated by this equation. The wave equation describing such a quantum entity is often called the wave function. De Broglie's discovery was enthusiastically endorsed by Einstein, and experiments in which the wavelengths of electrons were measured were carried out later in the 1920s. These showed that everything in the quantum realm is both particle and wave at the same time.

tennis ball arrives
at a definite place

ball spreads out into a wave form and goes through both holes at once

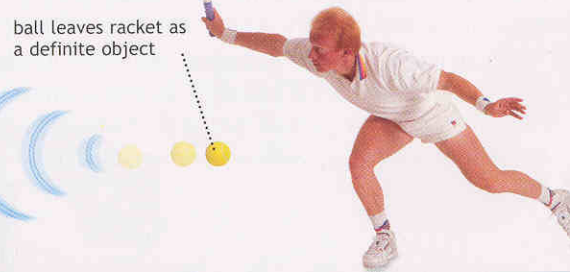
$\eta = 0.000000000000000000000000000066$

$m=0.00000000000000000000000000000099$

However, the quantum realm is limited to the world of the very small, because Planck's constant is so small. In units where mass is measured in grams, the value of h is 6.6×10^{-27} (a decimal point followed by 26 zeroes and two sixes). Since de Broglie's equation says that the wavelength of an object is equal to this tiny number divided by its momentum, and momentum is related to mass, the wavelength will only be detectable for objects with tiny masses. Sure enough, in the same units, the mass of an electron is 9×10^{-28} grams. Everything fits.

electron patterns

Although all this was clear to the experts before the end of the 1920s, the full drama of wave-particle duality and the probabilistic element that had been found in quantum physics was brought home in the 1980s by a team of Japanese scientists working at the Hitachi Research Laboratories. They carried out the double-slit experiment with electrons, which were fired, one at a time, through the equipment. The electrons were detected by a screen (like a TV screen) on the other side of the slits, with each electron making a spot of light when it arrived. The screen then “remembered” the spot, so that an image gradually built up as more and more electrons arrived.



ball leaves racket as
a definite object

size matters
Quantum effects are important for electrons because if you write the numbers out in the appropriate units, the mass (m) of an electron is even smaller than Planck's constant (h).

if h were larger
If Planck's constant
were big enough,
tennis balls would
behave like
electrons. After
you hit them, they
would dissolve into
waves as they
traveled from one
place to another,
and only condense
back into particles
when they hit
something.